Area 5: Mechanical Components and Assemblies

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

Contents

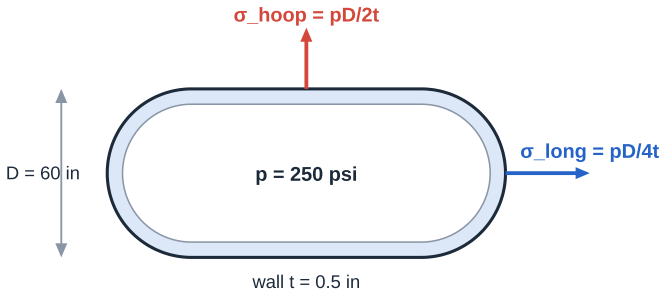
5.A Pressurized Vessels and Piping
5.B Hydraulic and Pneumatic Components
★ 5.C Beams, Trusses, and Frames (Statics)
5.D Springs
★ 5.E Vibrating Systems
★ 5.F Basic Machines and Mechanisms (Dynamics)

5.A Pressurized Vessels and Piping

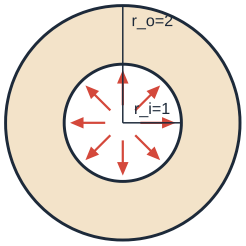
1. Key concepts

- Thin wall (t < D/20): hoop σ_t = pD/(2t), longitudinal σ_L = pD/(4t). Sphere: σ = pD/(4t).
- Thick wall (Lamé): σ_t = [p_ir_i² − p_or_o² + r_i²r_o²(p_i − p_o)/r²]/(r_o² − r_i²). Internal pressure only: the maximum is at r_i, σ_t = p(r_o² + r_i²)/(r_o² − r_i²).
- ASME VIII Div. 1 shell: t = PR/(SE − 0.6P), with R the inside radius and E the joint efficiency.

Diagrams



Thin-wall cylinder: hoop = pD/2t, longitudinal = pD/4t.



p_i = 23,000 psi

p_o = 0

Lamé: σ_t max at the bore

When t > r/10, use the Lamé thick-wall equations.

2. Essential formulas

Name	Formula	Variables / units
Hoop	$\sigma_t = pD/(2t)$	
Longitudinal	$\sigma_L = pD/(4t)$	
Lamé (internal)	$\sigma_{t,max} = p(r_o^2 + r_i^2)/(r_o^2 - r_i^2)$	
ASME shell	$t = PR/(SE - 0.6P)$	

3. Handbook search keywords

thin-walled pressure vessel | Lamé thick-walled cylinder | ASME Section VIII thickness

4. Worked examples

Example 1: ASME shell thickness

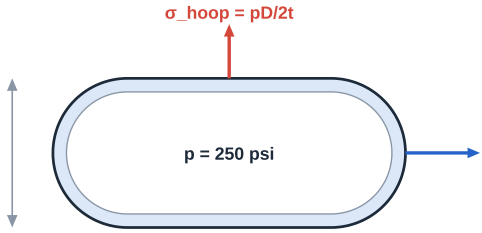
R = 30 in, P = 250 psi, S = 20,000 psi, E = 0.85. Find t.

Handbook search: **ASME Section VIII**

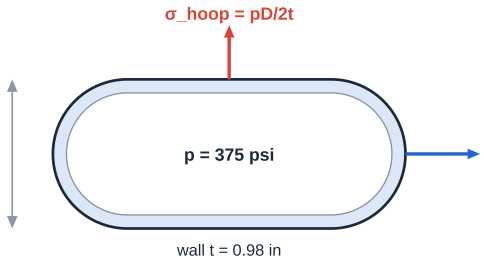
1. t = 250(30)/(20,000·0.85 − 0.6·250) = 0.445 in (add the corrosion allowance).

Answer: 0.445 in

Check your answer: t should come out close to the thin-wall PR/(SE), just slightly larger.



Try it yourself: ASME shell thickness
R = 36 in, P = 375 psi, S = 20,000 psi, E = 0.7. Find t (circumferential).
Handbook search: **ASME Section VIII**

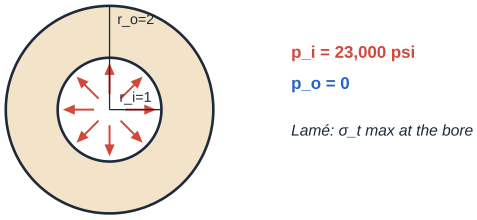


1. t = PR/(SE − 0.6P) = 375 × 36/(14,000 − 225) = 0.980 in.
2. Add corrosion allowance if specified, then pick the next plate size.

Answer: 0.980 in

Check your answer: Check that t < R/2 and P < 0.385SE, so the formula applies.

Example 3: Thick wall bore stress
r_i = 1 in, r_o = 2 in, p_i = 23,000 psi. Find the max hoop stress.
Handbook search: **Lamé thick wall**



1. $\sigma_t = p(r_o^2 + r_i^2)/(r_o^2 - r_i^2) = 23{,}000 \times 5/3 = 38{,}333$ psi.

Answer: 38,333 psi

Check your answer: At the bore, the radial stress is −p, and the hoop stress is always larger than p.

5. Common traps

- Using the radius in pD/2t (D is the diameter).
- Using the thin-wall formula when t/r > 0.1.

5.A Pressurized Vessels and Piping: one-page cheat summary

► $\sigma_t = pD/2t$; $\sigma_L = pD/4t$

► Lamé max at the bore

► $t = PR/(SE - 0.6P)$

Formulas

Hoop	$\sigma_t = pD/(2t)$
Longitudinal	$\sigma_L = pD/(4t)$
Lamé (internal)	$\sigma_{t,max} = p(r_o^2 + r_i^2)/(r_o^2 - r_i^2)$
ASME shell	$t = PR/(SE - 0.6P)$

Search keywords

thin-walled pressure vessel, Lamé thick-walled cylinder, ASME Section VIII thickness

Top traps

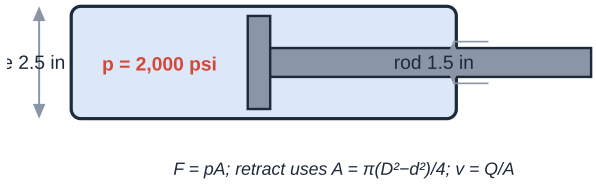
- Using the radius in pD/2t (D is the diameter).
- Using the thin-wall formula when t/r > 0.1.

5.B Hydraulic and Pneumatic Components

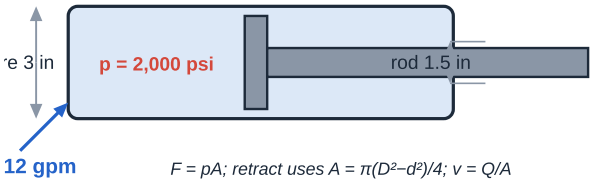
1. Key concepts

- Cylinder force F = pA. On retraction use the annulus A = π(D² − d²)/4. Speed v = Q/A.
- Hydraulic power: hp = Q(gpm)·p(psi)/1,714. Input = output/η.
- Pneumatics: the air is compressible, so use absolute pressure in gas-law calculations.

Diagrams



Hydraulic cylinder: cap end vs rod end.



Speed from flow: v = Q/A.

2. Essential formulas

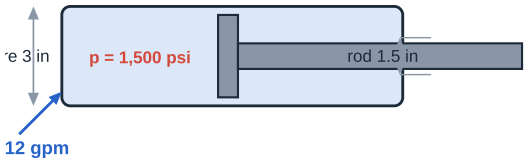
Name	Formula	Variables / units
Force	$F = pA$	
Speed	$v = Q/A\ (231\ \text{in}^3/\text{gal})$	
Power	$hp = Q{\cdot}p/1{,}714$	

3. Handbook search keywords

hydraulic cylinder force | hydraulic power 1714 | pump efficiency

4. Worked examples

Example 1: Cylinder speed and force
A Ø3 in bore cylinder at 1,500 psi is supplied with 12 gpm. Find the extension force and speed.
Handbook search: **hydraulic cylinder**

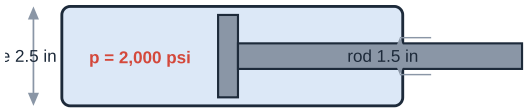


1. A = 7.069 in²; F = 10,603 lb.
2. v = 12(231)/60/7.069 = 6.54 in/s.

Answer: F ≈ 10,603 lb; v ≈ 6.54 in/s

Check your answer: 1 gal = 231 in³. Extension is slower and stronger than retraction.

Try it yourself: Retraction force
2.5 in bore, 1.5 in rod, 2,000 psi. Find the retraction force.
Handbook search: **hydraulic cylinder**



1. Annulus area = π(2.5² − 1.5²)/4 = 3.142 in².
2. F = pA = 6,283 lb.

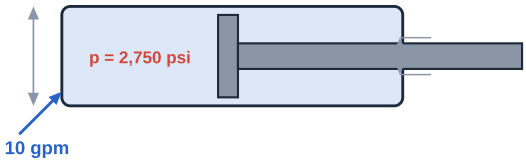
Answer: 6,283 lb

F = pA; retract uses A = π(D²−d²)/4; v = Q/A

Check your answer: Must be less than the extension force 9,817 lb.

Example 3: Pump input power
A pump delivers 10 gpm at 2,750 psi with η = 0.85. Find the input hp.

Handbook search: **hydraulic horsepower**



1. Hydraulic hp = Qp/1714 = 16.04 hp.
2. Input = 16.04/0.85 = 18.88 hp.

Answer: 18.9 hp

Check your answer: Input power must exceed the hydraulic output.

5. Common traps

- Using the full bore area on retraction.
- Gauge pressure in gas laws.

5.B Hydraulic and Pneumatic Components: one-page cheat summary

- ► $F = pA$
- ► $v = Q/A$
- ► $hp = Qp/1{,}714$

Formulas

Force	$F = pA$
Speed	$v = Q/A$ $(231\ in^3/gal)$
Power	$hp = Q{\cdot}p/1{,}714$

Search keywords

hydraulic cylinder force, hydraulic power 1714, pump efficiency

Top traps

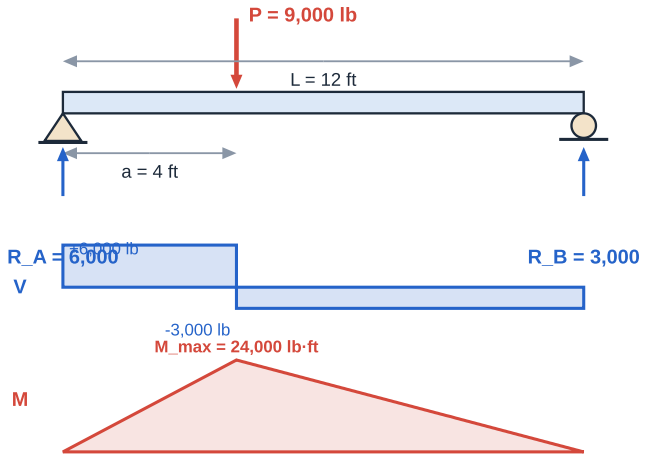
- Using the full bore area on retraction.
- Gauge pressure in gas laws.

5.C Beams, Trusses, and Frames (Statics)

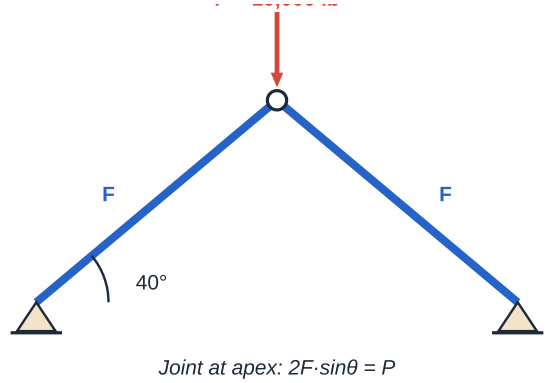
1. Key concepts

- Free-body diagram first: isolate the body, show every external force and moment (weight, supports, friction). Unknown reactions: pin = 2 forces, roller = 1, fixed = 2 forces + 1 moment.
- Equilibrium: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$. Take moments about the point with the most unknowns.
- Trusses: method of joints ($\Sigma F = 0$ at each pin; start where there are ≤ 2 unknowns) or method of sections (cut ≤ 3 members, then take moments). Zero-force members: at an unloaded joint where two members are collinear, the third member is zero.
- Friction: $F \le \mu_s N$. Impending motion means $F = \mu_s N$. Belt/capstan: $T_1/T_2 = e^{\mu\theta}$.
- Centroids and moments of inertia of composites: $\bar{x} = \Sigma A x / \Sigma A$, then $I = \Sigma(I_c + Ad^2)$.

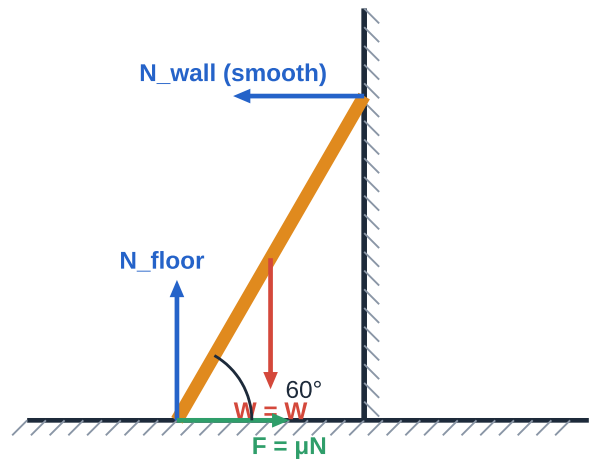
Diagrams



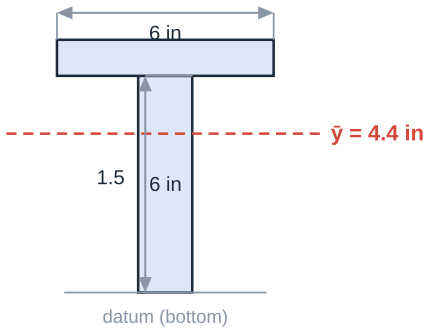
Beam reactions from ΣM = 0.



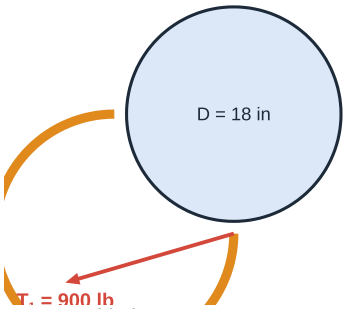
Method of joints at the apex.



Ladder FBD: smooth wall, rough floor.



Composite-area centroid.



Capstan friction on a post.

2. Essential formulas

Name	Formula	Variables / units
Equilibrium	$\Sigma F = 0 \; ; \Sigma M = 0$	
Friction	$F = \mu N$ (impending)	
Capstan	$T_1 = T_2 e^{\wedge}(\mu\theta)$	θ rad
Centroid	$\bar{y} = \Sigma A_i y_i / \Sigma A_i$	
Parallel axis	$I = \Sigma(I_c + Ad^2)$	
Two-force member	force along the line joining its pins	

3. Handbook search keywords

equilibrium · method of joints · method of sections · zero-force member · friction · centroid · parallel axis

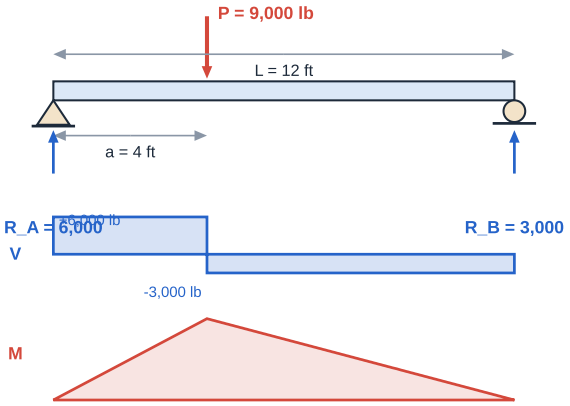
4. Worked examples

Example 1: Beam reaction
A 12 ft simple beam carries 9,000 lb at 4 ft from support A. Find R_A and R_B.
Handbook search: **equilibrium**

1. ΣM_B = 0: R_A(12) = 9,000(8).
2. R_A = 6,000 lb; R_B = 9,000 − 6,000 = 3,000 lb.

Answer: R_A = 6,000 lb, R_B = 3,000 lb

Check your answer: R_A + R_B must equal 9,000 lb.

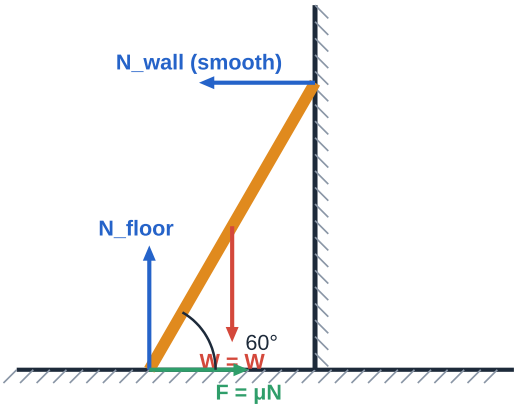


Example 2: Ladder friction
A uniform ladder leans at 60° from the floor against a smooth wall. Find the minimum μ at the floor.
Handbook search: **friction**

1. Smooth wall: horizontal reaction N_w only. Floor: N = W, F = N_w.
2. ΣM about the foot: N_w·L sin60 = W(L/2)cos60 → N_w = W/(2 tan60).
3. μ_min = N_w/N = 1/(2 tan60°) = 0.289.

Answer: 0.289

Check your answer: A steeper ladder needs less friction; at 90° it needs none.

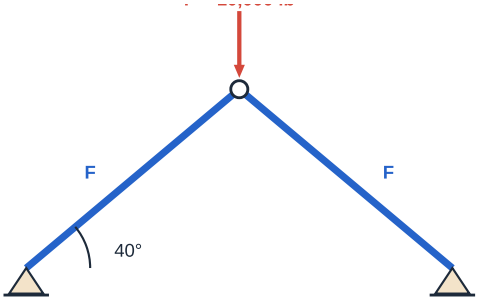


Example 3: Truss member
Two members meet at an apex at 40° to the horizontal and carry 10,000 lb vertically. Find the force in each.
Handbook search: **method of joints**

1. 2F sin40° = 10,000 → F = 7,779 lb (compression).

Answer: 7,779 lb

Check your answer: The members are in compression (they push on the joint) for a load hanging below a peaked frame.



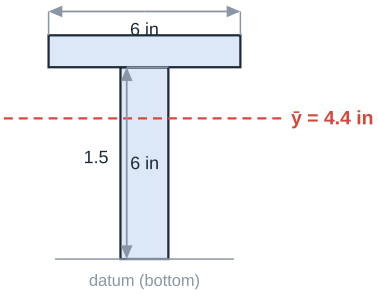
Joint at apex: 2F·sinθ = P

Try it yourself: T-section centroid
Flange 6 × 1 in on a web 1.5 × 6 in. Find ȳ from the bottom.
Handbook search: **centroid composite area**

1. A_1 = 6 in² at y = 6.5; A_2 = 9 in² at y = 3.
2. ȳ = ΣAy/ΣA = (39 + 27)/15 = 4.40 in.

Answer: 4.40 in

Check your answer: ȳ must fall between the two part centroids (3 and 6.5 in), nearer the larger area.

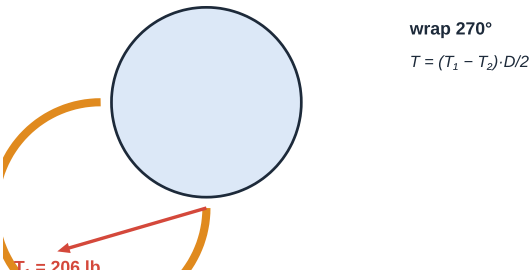


Example 5: Capstan friction
A rope is wrapped 270° around a post (μ = 0.3), with 50 lb on the slack side. Find the max tight-side tension.
Handbook search: **belt friction**

1. θ = 270° = 4.712 rad.
2. $T_1 = T_2 e^{(\mu\theta)}$ = 50e^(1.414) = 206 lb.

Answer: 206 lb

Check your answer: Each extra wrap multiplies the holding force. Doubling θ squares the ratio.



5. Common traps

- Leaving the friction force or the weight off the FBD.
- Adding a 'ma' force to a static FBD.

- Using cos where sin belongs (draw the angle).
- Taking moments about a point where the unknowns do not drop out.

5.C Beams, Trusses, and Frames (Statics): one-page cheat summary

- ▶ FBD → $\Sigma F = 0, \Sigma M = 0$
- ▶ Pin 2, roller 1, fixed 3 unknowns
- ▶ Zero-force: 2 collinear + 1 at an unloaded joint
- ▶ $F = \mu N$ impending
- ▶ $T_1/T_2 = e^{\wedge}(\mu\theta)$
- ▶ $\bar{y} = \Sigma Ay/\Sigma A$; $I = \Sigma(I_c + Ad^2)$

Formulas

Equilibrium	$\Sigma F = 0$; $\Sigma M = 0$
Friction	$F = \mu N$ (impending)
Capstan	$T_1 = T_2 e^{\wedge}(\mu\theta)$
Centroid	$\bar{y} = \Sigma A_i y_i / \Sigma A_i$
Parallel axis	$I = \Sigma(I_c + Ad^2)$
Two-force member	force along the line joining its pins

Search keywords

equilibrium, method of joints, method of sections, zero-force member, friction, centroid, parallel axis

Top traps

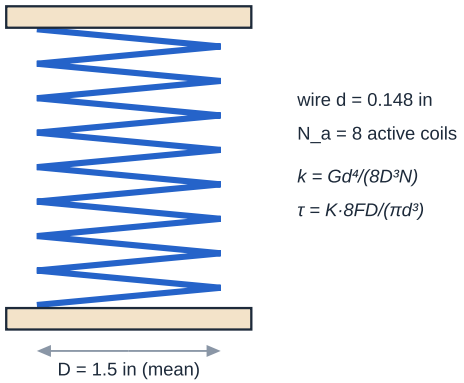
- Leaving the friction force or the weight off the FBD.
- Adding a 'ma' force to a static FBD.
- Using cos where sin belongs (draw the angle).

5.D Springs

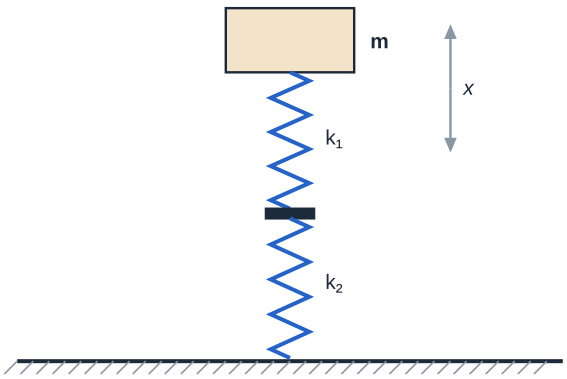
1. Key concepts

- Helical compression spring: k = Gd⁴/(8D³N_a). Spring index C = D/d (4–12 is good).
- Stress: τ = K_W·8FD/(πd³), with Wahl K_W = (4C − 1)/(4C − 4) + 0.615/C (or Bergsträsser K_B = (4C + 2)/(4C − 3)).
- Series springs: 1/k = Σ1/kᵢ. Parallel: k = Σkᵢ. Energy U = ½kx².
- Spring fatigue: Zimmerli data with torsional Goodman. Surge frequency: avoid resonance with the operating frequency.

Diagrams



Helical spring geometry: d, D, N.



Springs in series and parallel.

2. Essential formulas

Name	Formula	Variables / units
Rate	$k = Gd^4/(8D^3N)$	
Stress	$\tau = K{\cdot}8FD/(\pi d^3)$	
Wahl	$K = (4C-1)/(4C-4) + 0.615/C$	
Series/parallel	$1/k = \Sigma 1/k_i$; $k = \Sigma k_i$	

3. Handbook search keywords

helical spring rate · Wahl factor · spring index · springs in series

4. Worked examples

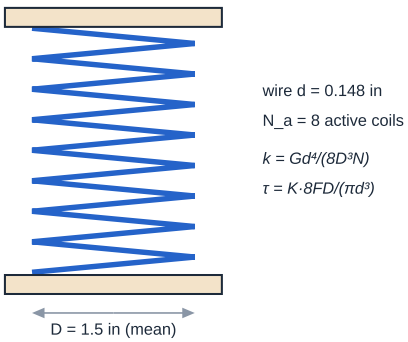
Example 1: Spring rate
Music wire G = 11.5×10⁶ psi, d = 0.148 in, D = 1.5 in, N_a = 8. Find k.

Handbook search: **spring rate**

1. k = 11.5×10⁶(0.148)⁴/(8·1.5³·8) = 25.5 lb/in.

Answer: 25.5 lb/in

Check your answer: k scales with d⁴ and 1/D³. Small changes in wire size matter a lot.



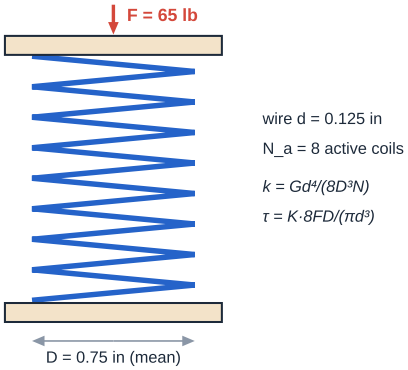
Try it yourself: Wahl-corrected stress
d = 0.125 in, D = 0.75 in, F = 65 lb. Find τ_max with the Wahl factor.

Handbook search: **Wahl factor**

1. C = D/d = 6.0.
2. K_W = (4C − 1)/(4C − 4) + 0.615/C = 1.2525.
3. τ = K_W·8FD/(πd³) = 79,609 psi.

Answer: 79,609 psi

Check your answer: C between 4 and 12 is normal. K_W is about 1.25 at C = 6.



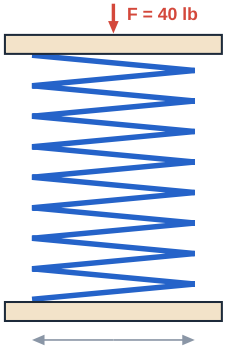
Example 3: Free length
k = 45 lb/in, working load 40 lb, solid height 1.0 in, 20% overrun to closure. Find the free length.

Handbook search: **free length**

1. Working deflection = 40/45 = 0.889 in.
2. Deflection to solid = 1.2 × 0.889 = 1.067 in.
3. L_free = L_s + y_s = 2.067 in.

Answer: 2.07 in

Check your answer: Free length must exceed solid height plus the working deflection.



5. Common traps

- Using the outside diameter for D (use the mean coil diameter).
- Leaving out the curvature factor.

5.D Springs: one-page cheat summary

- $k = Gd^4/8D^3N$
- $\tau = K8FD/\pi d^3$
- Series $1/k = \Sigma 1/k$

Formulas

Rate	$k = Gd^4/(8D^3N)$
Stress	$\tau = K{\cdot}8FD/(\pi d^3)$
Wahl	$K = (4C - 1)/(4C - 4) + 0.615/C$
Series/parallel	$1/k = \Sigma 1/k_i$; $k = \Sigma k_i$

Search keywords

helical spring rate, Wahl factor, spring index, springs in series

Top traps

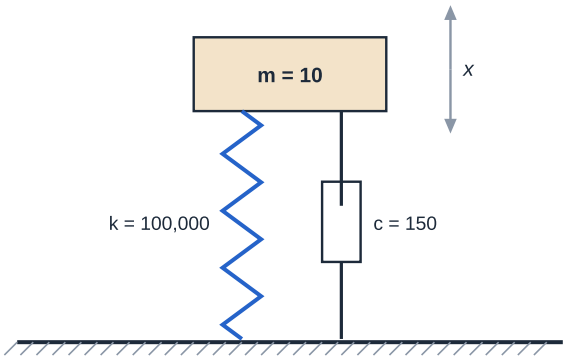
- Using the outside diameter for D (use the mean coil diameter).
- Leaving out the curvature factor.

5.E Vibrating Systems

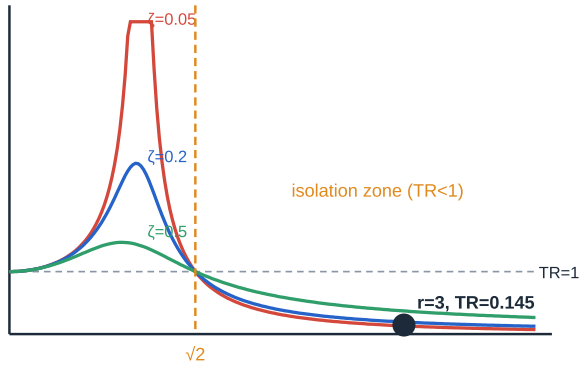
1. Key concepts

- SDOF: $\omega_n = \sqrt{(k/m)}$, $f_n = \omega_n/2\pi$. Static deflection shortcut: $\omega_n = \sqrt{(g/\delta_{st})}$, $f_n \approx 3.13/\sqrt{\delta}$(in) Hz.
- Damping: $c_c = 2\sqrt{(km)} = 2m\omega_n$, $\zeta = c/c_c$. Damped frequency $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$. Log decrement $\delta = (1/n)\ln(x_0/x_n)$, $\zeta = \delta/\sqrt{(4\pi^2 + \delta^2)}$.
- Forced response: $X = (F_0/k)/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$, with $r = \omega/\omega_n$. Resonance is near $r = 1$.
- Transmissibility $TR = \sqrt{[(1 + (2\zeta r)^2)/((1 - r^2)^2 + (2\zeta r)^2)]}$. Isolation ($TR < 1$) needs $r > \sqrt{2}$. Above $\sqrt{2}$, damping INCREASES TR, and softer springs or more mass help. Undamped: $TR = 1/(r^2 - 1)$.
- Rotating unbalance: $X = (m_e e/M)r^2/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$.
- Equivalent systems: springs in series or parallel. Cantilever $k = 3EI/L^3$; simply supported center $k = 48EI/L^3$. Torsional $k_t = GJ/L$, $\omega_n = \sqrt{(k_t/J_{mass})}$. Compound pendulum $\omega_n = \sqrt{(mgd/I_O)}$.

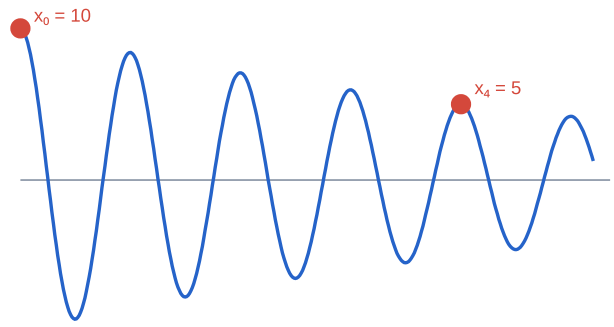
Diagrams



SDOF spring-mass-damper.

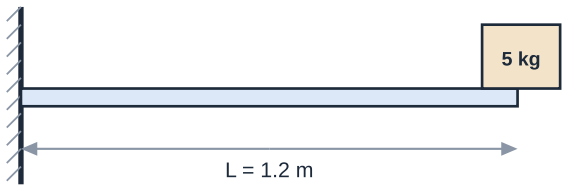


Transmissibility vs frequency ratio: isolation needs $r > \sqrt{2}$.



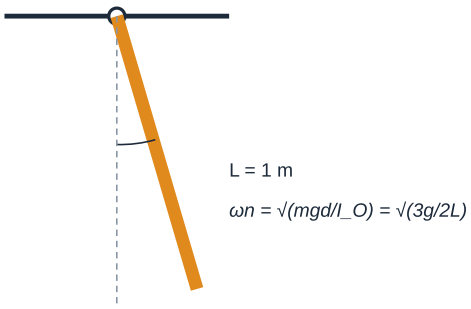
$\delta = (1/n)\ln(x_0/x_n)$, $n = 4$ cycles

Free decay: log decrement from successive peaks.



section 40 × 8 mm; $k = 3EI/L^3$

Mass at the tip of a cantilever.



Compound pendulum.

2. Essential formulas

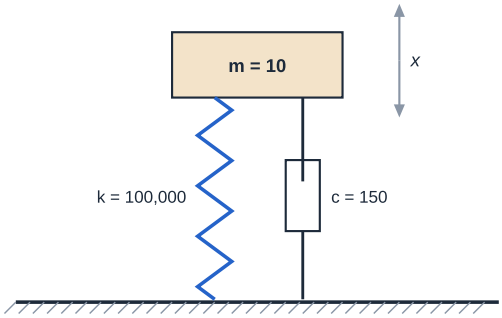
Name	Formula	Variables / units
Natural frequency	$\omega_n = \sqrt{(k/m)}$; $f = \omega_n/2\pi$	
Static deflection	$\omega_n = \sqrt{(g/\delta_{st})}$	
Damping ratio	$\zeta = c/(2\sqrt{(km)})$	
Damped	$\omega_d = \omega_n\sqrt{(1-\zeta^2)}$	
Log decrement	$\delta = (1/n)\ln(x_0/x_n)$; $\zeta \approx \delta/2\pi$	
Amplitude	$X = (F_0/k)/\sqrt{[(1-r^2)^2 + (2\zeta r)^2]}$	
Transmissibility	$TR = \sqrt{[(1+(2\zeta r)^2)/((1-r^2)^2 + (2\zeta r)^2)]}$	
Beam stiffness	cantilever $3EI/L^3$; SS center $48EI/L^3$	

3. Handbook search keywords

natural frequency · damping ratio · logarithmic decrement · transmissibility · vibration isolation · rotating unbalance · forced vibration

4. Worked examples

Example 1: Damping and frequency
m = 10 kg, k = 100 kN/m, c = 150 N·s/m. Find f_n, ζ and f_d.
Handbook search: **damping ratio**



1. ω_n = √(100,000/10) = 100.0 rad/s → f_n = 15.92 Hz.
2. c_c = 2√(km) = 2,000 N·s/m; ζ = 150/2,000 = 0.075.
3. f_d = f_n√(1 − ζ²) = 15.87 Hz (practically the same).

Answer: f_n = 15.92 Hz, ζ = 0.075

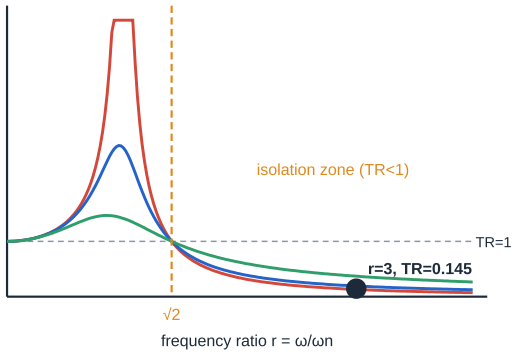
Check your answer: ζ < 1 means it oscillates. f_d ≈ f_n for light damping.

Example 2: Isolation
A machine on mounts runs at r = 3 with ζ = 0.1. Find TR. Would more damping help?
Handbook search: **transmissibility**

1. TR = √[(1 + 0.6²)/((1 − 9)² + 0.6²)] = 0.1454.
2. No. Above r = √2, extra damping raises TR.

Answer: TR ≈ 0.145

Check your answer: Above r = √2, more damping RAISES TR slightly.

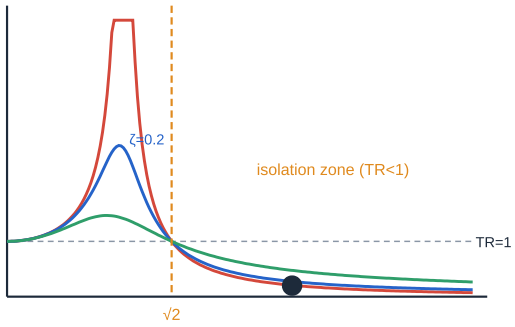


Example 3: Isolator static deflection
Undamped mounts must give TR = 0.20 at 1,800 rpm. Find the required static deflection.
Handbook search: **isolation static deflection**

1. r² = 1/TR + 1 = 6 → r = 2.449.
2. ω = 1,800·2π/60 = 188.5 rad/s; ω_n = ω/r = 77.0 rad/s.
3. δ = g/ω_n² = 386.1/5922 = 0.0652 in.

Answer: 0.065 in

Check your answer: A lower TR target needs a softer mount, meaning a larger static deflection.

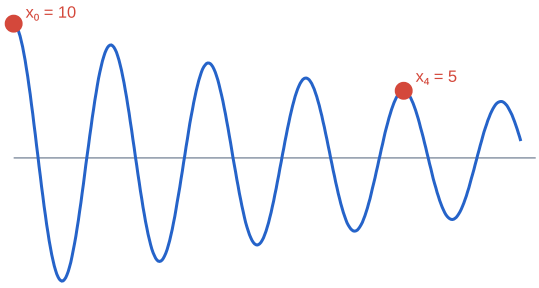


Try it yourself: Log decrement
Peak 10 mm, and 4 cycles later 5 mm. Find ζ.
Handbook search: **logarithmic decrement**

1. δ = (1/n)ln(x_0/x_n) = 0.1733.
2. ζ = δ/√(4π² + δ²) = 0.0276.

Answer: ζ ≈ 0.028

Check your answer: For small δ, ζ ≈ δ/2π = 0.0276. Nearly the same.



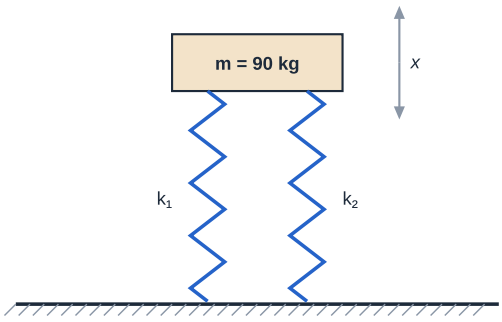
δ = (1/n)ln(x_0/x_n), n = 4 cycles

Example 5: Springs in parallel
A 90 kg mass on k_1 = 30,000 and k_2 = 10,000 N/m in parallel. Find f_n.
Handbook search: **natural frequency**

1. k_eq = k_1 + k_2 = 40,000 N/m.
2. f_n = (1/2π)√(k/m) = 3.36 Hz.

Answer: 3.36 Hz

Check your answer: Parallel springs are stiffer than either alone, so f_n is higher.



5. Common traps

- Mixing Hz and rad/s (ω = 2πf).

- Thinking damping always helps isolation (it hurts above √2).
- Leaving k in kN/m or N/mm while m is in kg.
- Using the series formula for parallel springs.

5.E Vibrating Systems: one-page cheat summary

- ► $\omega_n = \sqrt{(k/m)} = \sqrt{(g/\delta)}$
- ► $\zeta = c/2\sqrt{(km)}$
- ► $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$
- ► $\zeta \approx \ln(x_0/x_n)/(2\pi n)$
- ► $TR < 1$ only for $r > \sqrt{2}$
- ► Undamped $TR = 1/(r^2 - 1)$
- ► k: cantilever $3EI/L^3$, SS $48EI/L^3$

Formulas

Natural frequency	$\omega_n = \sqrt{(k/m)}$; $f = \omega_n/2\pi$
Static deflection	$\omega_n = \sqrt{(g/\delta_{st})}$
Damping ratio	$\zeta = c/(2\sqrt{(km)})$
Damped	$\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$
Log decrement	$\delta = (1/n)\ln(x_0/x_n)$; $\zeta \approx \delta/2\pi$
Amplitude	$X = (F_0/k)/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$
Transmissibility	$TR = \sqrt{[(1 + (2\zeta r)^2)/((1 - r^2)^2 + (2\zeta r)^2)]}$
Beam stiffness	cantilever $3EI/L^3$; SS center $48EI/L^3$

Search keywords

natural frequency, damping ratio, logarithmic decrement, transmissibility, vibration isolation, rotating unbalance, forced vibration

Top traps

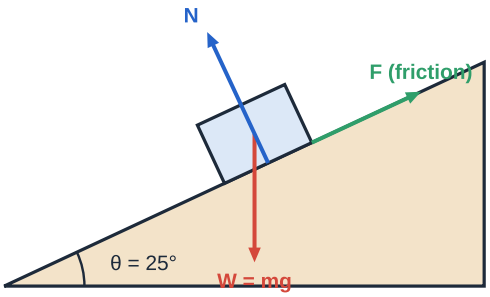
- Mixing Hz and rad/s ($\omega = 2\pi f$).
- Thinking damping always helps isolation (it hurts above $\sqrt{2}$).
- Leaving k in kN/m or N/mm while m is in kg.

5.F Basic Machines and Mechanisms (Dynamics)

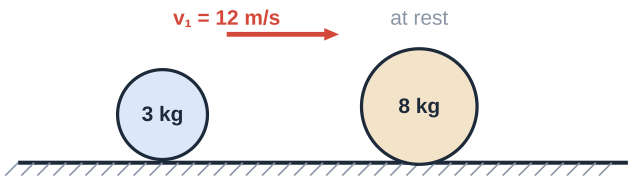
1. Key concepts

- Kinematics: constant acceleration gives $v = v_0 + at$, $s = v_0t + \tfrac{1}{2}at^2$, $v^2 = v_0^2 + 2as$. Rotation is the same with θ, ω, α, and $v = r\omega$, $a_t = r\alpha$, $a_n = r\omega^2$.
- Kinetics: $\Sigma F = ma$ and $\Sigma M_G = I_G\alpha$. Draw the FBD and the kinetic diagram.
- Work–energy: $T_1 + \Sigma U = T_2$, with $T = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2$. Spring energy $\tfrac{1}{2}kx^2$, gravity mgh, friction work μNd.
- Impulse–momentum: $mv_1 + \int F\,dt = mv_2$. Impact: momentum is conserved; $e = (v_2' - v_1')/(v_1 - v_2)$.
- Rolling without slip: $a = g\sin\theta/(1 + I/(mr^2))$. Hoop: $I = mr^2$. Solid disk: $\tfrac{1}{2}mr^2$. Sphere: $0.4mr^2$.
- Mechanisms: mechanical advantage, linkages (Grashof), cams, and gear/belt speed ratios.

Diagrams

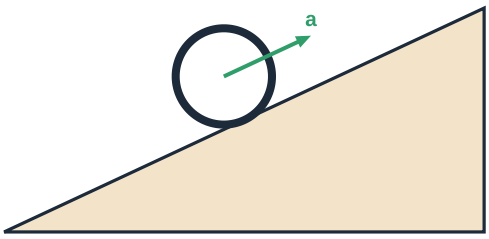


Block on an incline: weight components, normal, friction.



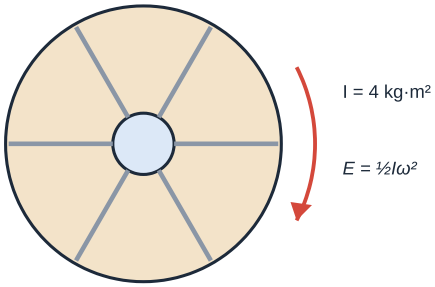
Momentum conserved; e = 0.8

Direct central impact with restitution.



Rolling hoop: a = g·sinθ/(1 + I/mr²)

Rolling without slip on an incline.



Rotor spin-up: T = Iα.

2. Essential formulas

Name	Formula	Variables / units
Kinematics	$v^2 = v_0^2 + 2as\ ;\ s = v_0t + \tfrac{1}{2}at^2$	
Newton	$\Sigma F = ma\ ;\ \Sigma M = I\alpha$	
Work–energy	$\tfrac{1}{2}mv_1^2 + \Sigma U = \tfrac{1}{2}mv_2^2$	
Restitution	$e = (v_2' - v_1')/(v_1 - v_2)$	
Rolling	$a = g\sin\theta/(1 + k^2/r^2)$	
Centripetal	$F = mr\omega^2 = mv^2/r$	

Braking distance	$d = v^2/(2\mu g)$	

3. Handbook search keywords

kinematics · Newton's second law · work-energy · impulse momentum · coefficient of restitution · rolling without slipping · mass moment of inertia

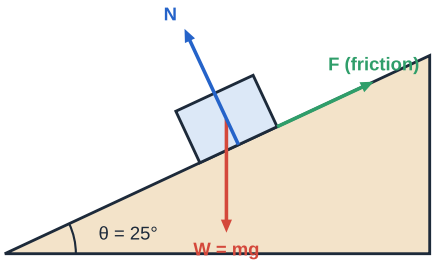
4. Worked examples

Example 1: Crate on an incline
A crate is released on a 25° incline with μ_k = 0.15. Find a and the speed after 4 m.
Handbook search: **Newton's second law**

1. a = g(sin25° − 0.15cos25°) = 2.81 m/s².
2. v = √(2as) = √(2·2.81·4) = 4.74 m/s.

Answer: a = 2.81 m/s², v = 4.74 m/s

Check your answer: a must be less than g·sinθ = 4.15 m/s² because friction opposes the motion.

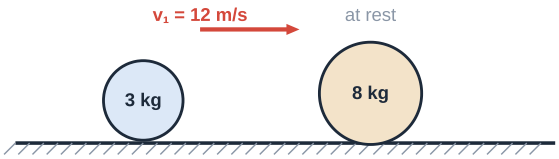


Example 2: Impact
A 3 kg ball at 12 m/s hits a stationary 8 kg ball, e = 0.8. Find v_2'.
Handbook search: **coefficient of restitution**

1. Momentum: 36 = 3v_1' + 8v_2'. Restitution: $v_2' - v_1'$ = 9.6.
2. $v_2' = m_1 v_1(1 + e)/(m_1 + m_2)$ = 3(12)(1.8)/11 = 5.89 m/s.

Answer: 5.89 m/s

Check your answer: Momentum before = after: 3 × 12 = 36 kg·m/s. Check that 3v_1' + 8v_2' = 36.



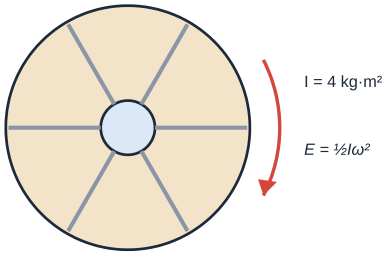
Momentum conserved; e = 0.8

Example 3: Spinning up a rotor
I = 4 kg·m², net torque 50 N·m. Time to reach 1,200 rpm?
Handbook search: **rotational dynamics**

1. ω = 1,200·2π/60 = 125.7 rad/s; α = T/I = 12.5 rad/s².
2. t = ω/α = 10.05 s.

Answer: 10.1 s

Check your answer: Using rad/s: ω = 125.7 rad/s, so t = Iω/T ≈ 10 s.

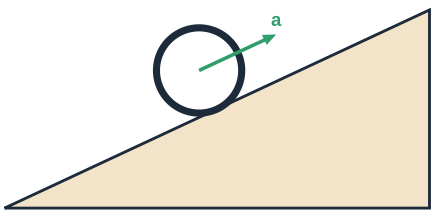


Try it yourself: Rolling hoop
A thin hoop rolls without slip down a 25° incline. Find a.
Handbook search: **rolling without slipping**

1. For a hoop I = mr², so a = g sinθ/(1 + I/mr²) = g sinθ/2.
2. a = 9.81 × sin25°/2 = 2.073 m/s².

Answer: 2.07 m/s²

Check your answer: A solid disk would be faster (2/3 g sinθ). A sliding block with no friction is faster still.



Rolling hoop: a = g·sinθ/(1 + I/mr²)

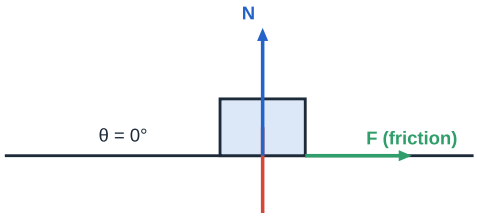
Example 5: Braking distance
40 km/h, wheels locked, μ = 0.7. Find the stopping distance.

Handbook search: **work-energy**

1. v = 40/3.6 = 11.11 m/s; a = μg = 6.87 m/s².
2. d = v²/(2a) = 8.99 m.

Answer: 9.0 m

Check your answer: Distance scales with v². At 80 km/h it would be 4× as far.



5. Common traps

- Forgetting the rotational KE of rolling bodies.
- Treating rpm as rad/s.
- Assuming energy is conserved in an impact (only momentum is).
- Getting the friction direction wrong (it opposes relative motion).

5.F Basic Machines and Mechanisms (Dynamics): one-page cheat summary

- ► $v^2 = v_0^2 + 2as$
- ► $\Sigma F = ma$, $\Sigma M = I\alpha$
- ► $T = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$
- ► e = sep/approach
- ► Rolling $a = g\sin\theta/(1 + I/mr^2)$
- ► $\omega = 2\pi n/60$

Formulas

Kinematics	$v^2 = v_0^2 + 2as$; $s = v_0t + \frac{1}{2}at^2$
Newton	$\Sigma F = ma$; $\Sigma M = I\alpha$
Work–energy	$\frac{1}{2}mv_1^2 + \Sigma U = \frac{1}{2}mv_2^2$
Restitution	$e = (v_2' - v_1')/(v_1 - v_2)$
Rolling	$a = g\sin\theta/(1 + k^2/r^2)$
Centripetal	$F = mr\omega^2 = mv^2/r$
Braking distance	$d = v^2/(2\mu g)$

Search keywords

kinematics, Newton's second law, work-energy, impulse momentum, coefficient of restitution, rolling without slipping, mass moment of inertia

Top traps

- Forgetting the rotational KE of rolling bodies.
- Treating rpm as rad/s.
- Assuming energy is conserved in an impact (only momentum is).